CO -5

Build a Flutter application with a user interface containing two textfields (for Name and Roll No) and a button. When the button is pressed, the data from the fields should be saved as a new document in a Firestore collection named students.

Aim

To develop a Flutter application with two text fields for entering **Name** and **Roll No**, and a button to store the entered details as a new document in a Firestore collection named **students**.

Requirements

- Flutter SDK
- Dart SDK
- Firebase
- Cloud Firestore
- Firebase CLI
- FlutterFire CLI
- VS Code / Android Studio

Theory

Firebase Cloud Firestore is a NoSQL cloud database that stores data in the form of collections and documents.

In this application, the student details are stored in the following structure:

```
students
   |
   └── Document ID
         |
         ├── name
         └── rollNo
```

The cloud_firestore package is used to connect the Flutter application with Firestore.

The add() method is used to create a new document with an automatically generated document ID.

Procedure

1. Create a new Flutter project.
2. Create a Firebase project.
3. Enable Cloud Firestore in the Firebase project.
4. Configure Firestore for the Flutter application.
5. Add the required Firebase packages using:

flutter pub add firebase_core

flutter pub add cloud_firestore

6. Configure Firebase using FlutterFire CLI.
7. Import the required Firebase packages into main.dart.
8. Initialize Firebase using Firebase.initializeApp().
9. Create two TextField widgets for entering Name and Roll No.
10. Create an ElevatedButton to save the student details.
11. When the button is pressed, retrieve the values from the text fields.
12. Store the values in the students Firestore collection using the add() method.

Program

```
import 'package:flutter/material.dart';

import 'package:firebase_core/firebase_core.dart';

import 'package:cloud_firestore/cloud_firestore.dart';

import 'firebase_options.dart';


void main() async {
```

```
  WidgetsFlutterBinding.ensureInitialized();

  await Firebase.initializeApp(
    options: DefaultFirebaseOptions.currentPlatform,
  );

  runApp(const MyApp());
}

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,
      title: 'Student Database',
      home: const StudentPage(),
    );
  }
}

class StudentPage extends StatefulWidget {
  const StudentPage({super.key});

  @override
```

```
  State<StudentPage> createState() => _StudentPageState();
}

class _StudentPageState extends State<StudentPage> {
  final TextEditingController nameController =
      TextEditingController();

  final TextEditingController rollNoController =
      TextEditingController();

  Future<void> saveStudent() async {
    String name = nameController.text.trim();
    String rollNo = rollNoController.text.trim();

    if (name.isEmpty || rollNo.isEmpty) {
      ScaffoldMessenger.of(context).showSnackBar(
        const SnackBar(
          content: Text('Please enter Name and Roll No'),
        ),
      );
      return;
    }

    await FirebaseFirestore.instance
        .collection('students')
        .add({
```

```
    'name': name,
    'rollNo': rollNo,
   });

   nameController.clear();
   rollNoController.clear();

   ScaffoldMessenger.of(context).showSnackBar(
    const SnackBar(
     content: Text('Student saved successfully!'),
    ),
   );
  }

 @override
 Widget build(BuildContext context) {
  return Scaffold(
   appBar: AppBar(
    title: const Text('Student Database'),
   ),
   body: Padding(
    padding: const EdgeInsets.all(20),
    child: Column(
     children: [
      TextField(
       controller: nameController,
```

```
          decoration: const InputDecoration(
            labelText: 'Name',
            border: OutlineInputBorder(),
          ),
        ),

        const SizedBox(height: 20),

        TextField(
          controller: rollNoController,
          decoration: const InputDecoration(
            labelText: 'Roll No',
            border: OutlineInputBorder(),
          ),
        ),

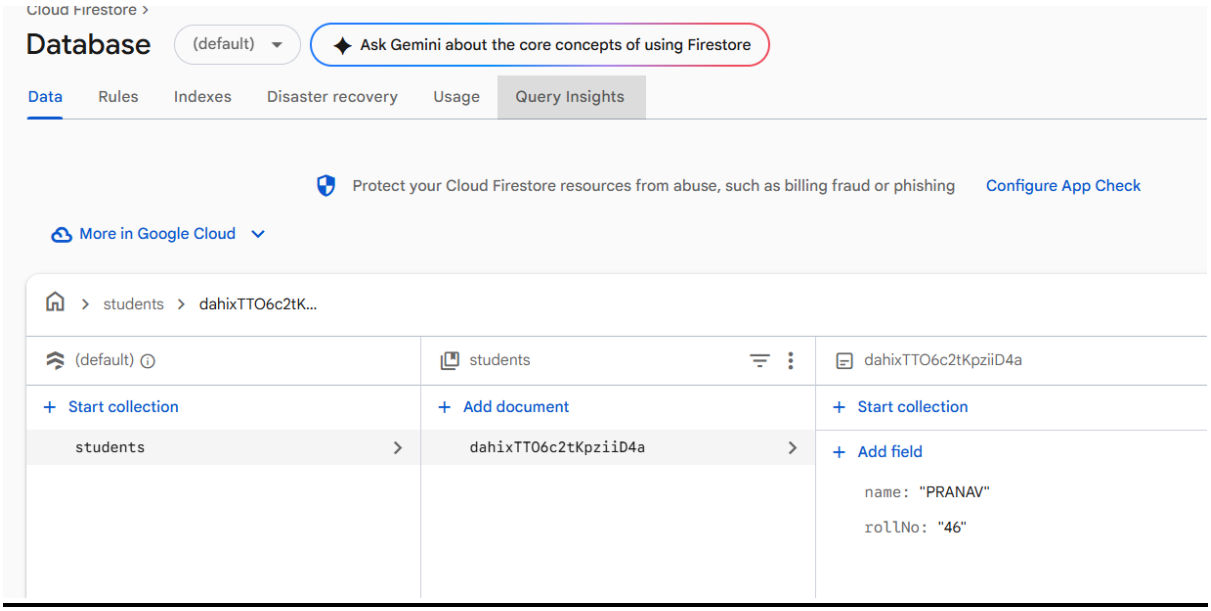
        const SizedBox(height: 20),

        ElevatedButton(
          onPressed: saveStudent,
          child: const Text('Save Student'),
        ),
      ],
    ),
  ),
);
```

```
  }
}
```

OUTPUT





Result

The Flutter application was successfully developed to accept student **Name** and **Roll No** and store the details as a new document in the **students** collection of Firebase Cloud Firestore.